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Refocusing Safety Metrics: Making Fatalities and Permanent Impairments the Benchmark for True Safety Performance

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Abstract

The process of switching over from TRIR to FPIR based system for safety performance evaluation is studied with respect to various safety parameters of different Oil and Gas industry majors such as ADNOC, Shell, Chevron, and Eni. The safety parameters studied include the Near Miss, Potential Incidents and injury rates in addition to fatalities. The awareness of keeping Health, Safety, Environment (HSE) over the operational goals of an organization slowly picked up momentum across the globe with due sustainability into consideration. The data with respect to TRIR and FPIR is studied and found that over the past decades, the oil and gas industry has made remarkable strides in improving occupational safety, notably reflected by a consistent decline in Total Recordable Injury Rates (TRIR). Data from the IOGP 2023f report demonstrates that TRIR has reached historically low levels across the upstream sector. However, the rate of Fatalities and Permanent Impairments (FPI) has remained a persistent concern. Analysis of ADNOC Group data between 2020 and 2024 similarly shows that while recordable incidents have significantly reduced, the occurrence of Fatalities and Permanent Impairments remains unstable. This paper argues that TRIR is no longer a sufficient proxy for safety performance excellence. Because the TRIR is based on the incidents counting without considering severity, it is estimated that TRIR may not be sufficient to address the safety scenario of performance of an Industry. Instead, industry benchmarking should shift towards FPI-focused metrics to drive the elimination of life-changing and life-ending events. Comparative analysis of industry and ADNOC trends is presented, culminating in a proposed data-driven framework to prioritize FPI as the primary safety performance benchmark. The industry level safety performance coupled with performance of ADNOC has proved that the drift towards FPIR will be more effective to assess the safety performance of an industry rather than TRIR based safety performance. This has been effectively demonstrated in this paper. In initial stage of switch over from TRIR to FPIR has some amount of reluctant adoption at various levels in oil and gas industry it has been gaining its strength to make FPIR as the basis for safety performance evaluation.

Introduction

The energy industry's steady reduction in Total Recordable Incident Rate (TRIR) over the past two decades has often been celebrated as a success story and is widely reported as evidence of continuous safety improvement. TRIR, which measures the frequency of recordable injuries per million manhours, became

a cornerstone benchmark for many International and National Oil Companies and trade associations such as International Oil and Gas Producers (IOGP) collates the data and produces yearly reports across the reporting member companies. This allows comparisons on safety performance across regions and companies, allowing organizations to demonstrate apparent progress toward "world-class" safety performance. Figure 1 below shows TRIR and FAR from 2000 till 2023 collate by IOGP.

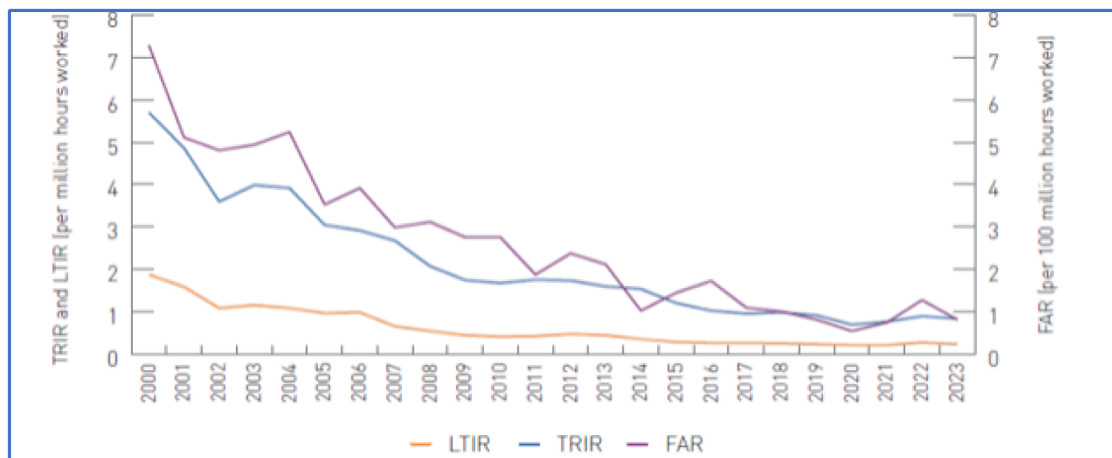


Figure 1—TRIR and FAR rates

As can be seen from Figure 1, despite decades of progress in reducing recordable incidents, safety performance across the energy industry has reached a plateau. Continued reliance on Total Recordable Injury Rate (TRIR) masks the true severity of incidents by giving equal weight to minor injuries and life-changing events like permanent impairments (PI). The industry benchmark reveals that while TRIR has improved, however, Fatalities Permanent Impairment (FPI) remains a concern. This disconnect underscores a critical need for more impactful metrics. Transitioning to Fatality Permanent Impairment will better align performance measurement with the outcomes that truly matter.

Over the years the utilization of safety metric, namely TRIR, has remained attractive for senior management to assess the safety performance, especially in Executive Leadership Meetings (ELT), Sustainability Reports, Regulatory Reports and purpose of investor communications (e.g. Merger and Acquisition). On the other hand, deep-dive analysis on Major Accidents across the industries, specifically on fatal incidents, have not indicated the same downward trend as TRIR. For example, high-profile major accidents such as Piper Alpha (1998), Texas City (2005), Buncefield (2005) and Macondo (2010), did exposed a critical flaw in relying on frequency-based metrics whereby while minor injuries decreased, rare but high-consequence incidents continued to occur with devastating people, environmental, asset and reputation damages. The disconnect between personal safety performance (e.g. low TRIR) and the persistent of catastrophic outcomes also indicates that frequency-based metrics may expose vulnerabilities in risk management systems. As matter of fact, by treating all incidents equally regardless of severity, TRIR can create a distorted picture of resilience and safety performance providing a false sense of assurance to senior management and true safety performance.

It is interesting to note that this challenge is not only unique to energy industry but across other high-hazard industries such as aviation, nuclear and others. Researchers and regulators have consistently debated the importance of weighting incidents by consequence rather than frequency alone. As highlighted earlier International Oil and Gas Producers (IOGP) via Project Safira, has acknowledge that TRIR, while useful metric to track minor injuries, is insufficient to drive the prevention of life-altering events namely permanent impairments. IOGP 2023 report emphasizes the need to adopt severity focused indicators such as FPI to align performance measures with what truly matters: Protecting Life and Avoiding Irreversible Harm.

The evolution of safety thinking further supports this shift. Traditional safety management has emphasized counting failures (lagging indicators), but contemporary approaches, such as Safety-II and resilience engineering, stress the need to understand the conditions that prevent catastrophic failure and to capture "how work actually happens." Within this paradigm, metrics must not only track the absence of harm but also provide meaningful insights into the presence of safety. Frequency based measures such as TRIR fall short in this regard compared to FPI which provides better reflection on safety performance as it focused approach addresses on elimination or minimization of severe consequences.

Against this backdrop, several leading companies, including ADNOC, Shell, and Chevron, have begun exploring FPI as a primary safety benchmark. Preliminary analyses across ADNOC Group between 2020 and 2024, for instance, reveal that while TRIR showed progressive declines, the occurrence of fatalities and permanent impairments remained unstable, underscoring the need for a different lens of evaluation. Moving toward FPI aligns with ADNOC's ambition to achieve 100% HSE and Zero Harm and positions the company as an industry leader in adopting safety performance measures that emphasize consequence, severity and resilience over frequency.

Research Objective & Hypothesis

This research seeks to critically evaluate the transition from TRIR to FPI as the benchmark for safety performance in the oil and gas industry, with specific emphasis on the ADNOC Group experience and its alignment with global industry trends. The objective is to determine whether FPI provides a more stable, severity-sensitive, and outcome-driven framework for assessing true safety performance.

The central hypothesis of this study is that FPI, as opposed to TRIR, offers a superior benchmark for safety excellence because it directly reflects the prevention of life-changing and irreversible harm. By capturing the severity dimension of incidents, FPI is expected to provide greater predictive validity for catastrophic risk management, enable more meaningful benchmarking across organizations, and strengthen leadership accountability for outcomes that matter most.

Statement of Theory

Health, Safety and Environment which affects the People, Environment, Assets and Reputation of an industry or an entity had become the focus of Energy Industry over decades. In this context there are developed many policies across the globe by various countries. The statutory policies along with voluntary absorption of maintaining a sustainable environment in Health and Safety as key points in the operations had contributed to the betterment of Energy industry over decades. Safety metrics in this regard required a revisit to understand the actual impact as per the data collected from various sources on safety performance indicators such as Fatality and Permanent Impairment Injuries (FPI) 2022-23. Notwithstanding IOGP Life Saving Rules (LSR) are collectively applied and standardized in Energy Industry to ensure safety.

The problem statement here is to refocus on the safety metrics to switch over to an FPI based benchmark for assessing the safety performance in the Energy Industry. It is to understand that TRIR was not effective in safety performance assessment as it gives the same weightage to all types of incidents without looking into the aspect of severity. Hence TRIR is not a proper safety performance indicator. For better assessment of safety performance, a severity base safety Risk Assessment Matrix (RAM) to be formed. It is based on FPIR. The transition from TRIR to FPIR based system requires a considerable effort from Industry players. The adaptation of TRIR for quite a long period already plays a role of perception while switching over to new system. There is a disconnect in the TRIR data and FPI data when analyzed with reference to Industry benchmark due to the approach adopted in both systems. This is since severity of incidents is not taken into consideration in the TRIR system.

In this research a deep-dive on safety performance based on FPI data, both downstream and upstream assets data was collected for various organizations such as ADNOC, Shell, Chevron, BP and Eni as these

companies are giants in Energy Industry across the Globe. The data of number of millions of man hours worked, number of serious injuries, number of fatalities, are collected for each company derived from annual report and sustainability reports. The FPI rate is calculated for each year as per the data for each company to understand the behavior of FPI rate pattern. This methodology is adopted in this paper to understand the effect of HSE performance using Fatalities and Permanent Injuries as the benchmarks for obtaining HSE performance of each company and to develop and safety metrics and as per it, to develop safety policies. Upon analyzing the data, the shift from frequency to severity (FPI) as per IOGP Recommended Practice is appreciated.

Fatality Permanent Impairment

The focus of the study in this paper is to understand the effect of shift from the TRIR to FPIR based safety indicators in actual safety performance. Safety performance on a continual basis has improved over decades, which is an undisputed fact in the Energy Industry. However, to know the ground level reality, a shift from the frequency-based reporting such as TRIR to Life Altering Events reporting to achieve the goal zero in harm and leaks such as goal no harm is necessitated and the same is implemented in the Energy Industry. This shift has improved safety performance as the goal post is shifted from frequency to severity. The TRIR is based on frequency whereas FPIR is based on severity. This perceptual and attitudinal shift had impacted positively on the performance of the Energy Industry over time. To study the adaptation and the effect of Severity based FPI system preference over frequency based TRIR system many equipment and processes are considered. Here the process of bench marking FPI over TRIR for assessing the safety performance over time of companies in Energy Industry using various equipment is discussed.

The process involves studying the data collected from various sources and processing the data for assessing the suitability of FPI as the benchmark for studying true safety performance. From industry side it involves upgrading of the industry performance from encouraging safety performance to eliminating hazards with various steps of bettering the HSE Gates scoring with hierarchy controls. The hierarchy of control involves steps starting from Encouraging HSE culture, educating about better HSE culture, Redesigning the process equipment for better HSE output, if required substituting the equipment, then proceeding towards final goal of eliminating the hazards. In the whole cycle if the end of eliminating hazards is achieved then it automatically involves in the achievement of Goal - No harm ([Reporting Centre and Archive | Sustainability | Home, n.d.](#))

The equipment or data resources used in the process are ADNOC processes, ADNOC classification, IOGP, Safety Performance indicators 2022-23 data with analysis on Fatal Accident Rate, Permanent Impairment Rate, FPIR, Potential Incident data, Cause and activity data, Life Saving Rules application data, Body parts injured data and Nature of injury data. The data collected for companies such as ADNOC, Shell, Chevron, and Eni also forms the equipment for assessing the FPI shift and its effect on safety performance to assess the true safety performance as benchmark.

Especially in ADNOC, the FPI metric was introduced in the year 2022 with a focus to prevent the most severe incidents. The data collected for the given year is compared between ADNOC and the IOCs in respect of FPIR. The HSE score card for various years is compared in respect of sustainability with various parameters of safety. The parameters include fatality, catastrophic events, TRIR, permanent impairment, and HIPO Near Miss. The study of FPI GATE scoring mechanism is adopted to understand the achievement towards goal zero.

FPI stimulation data for ADNOC Upstream and Downstream assets. The data is studied for effective implementation of FPI bases metric as per the GHSE Function assessment on each option. These options are weighed regularly and the process of migration to FPI based benchmark is studied for its effectiveness in safety performance of companies. The study of data in respect of TRIR and FPI based safety performance

parameters is done in this paper and the findings from above resources are discussed in the following paragraphs.

Presentation of Data and Results

The presentation of data is based on the data provide by IOGP 2023f report, for the duration from 2011-2023 as Industry TRIR Trends vs. FPIR trends ([Safety Performance Indicators - Fatality and Permanent Impairment Injuries \(FPI\) - 2022-2023 Data, 2022](#)). The other data regarding the ADNOC performance between 2022-24 is considered for comparing the TRIR Trends Vs. FPIR trends. In addition to the above, the ADNOC classification data from 2020-24 is also taken into consideration for study. The various data and analysis towards problem solving for switching to FPI based benchmarks are presented below.

Industry Context: IOGP 2023f Analysis

The IOGP 2023f report highlighted that while TRIR continues to fall, FPIR has shown minimal improvement after 2018 to till 2023. This indicates that efforts focused on minor injuries are not sufficient to prevent life-altering or fatal events.

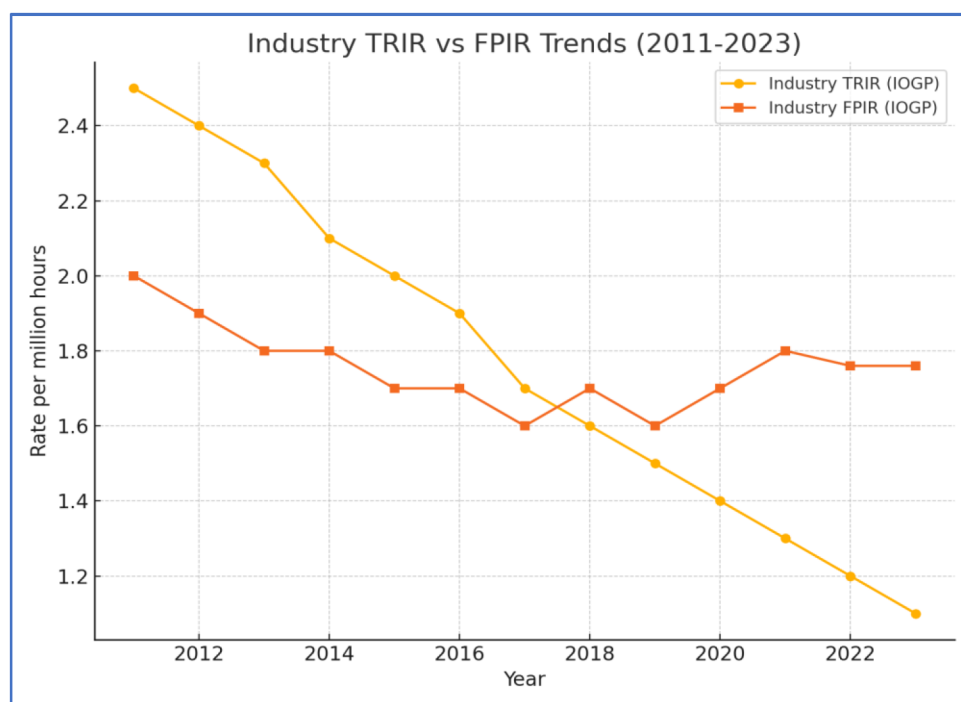


Figure 2—TRIR Vs. FPIR for Industry

Table 1—Year wise FPIR for various companies

FPIR for various companies' year wise			
Description	2022	2023	2024
ADNOC Group	1.40	0.90	0.62
Shell	2.00	2.60	1.50
Chevron	7.44	6.71	4.04
bp	1.74	0.96	1.46
Eni	2.46	0.63	4.96

The FPIR for the period from 2021-2024 as shown in Figure 3 below shows that all the major companies across the Globe indicated a fall Except Eni and Chevron in 2024 Shell PLC. (2023). This indicates the adoption issues in the new process of FPIR in some major companies which are expected to be in alignment with others in future.

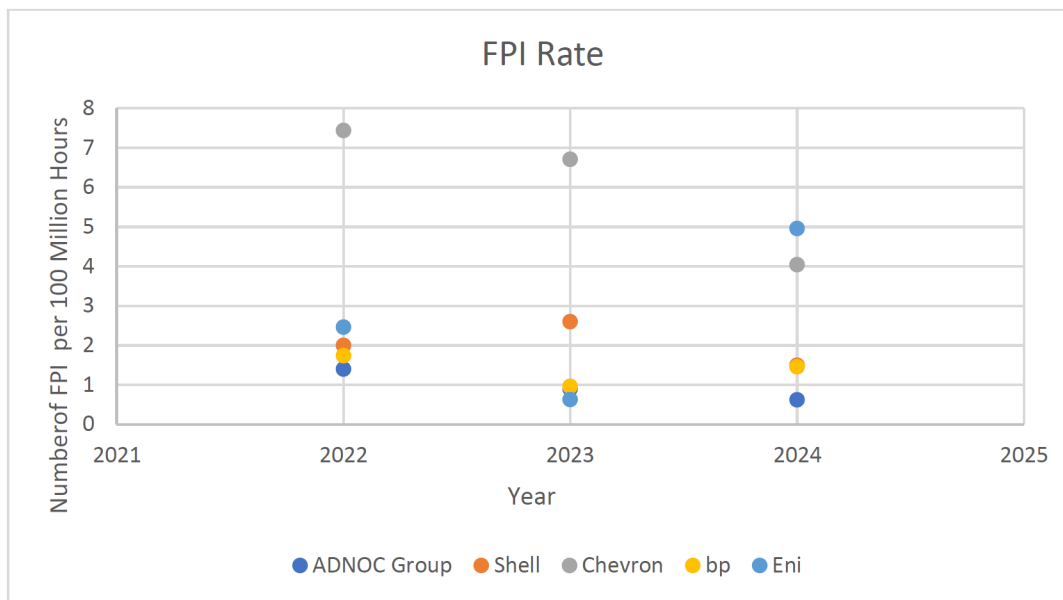


Figure 3—Company wise Year wise FPIR

In the above two figures, the Permanent Impairment (PI) is work related and FPIR is calculated as Number of Fatalities plus PI injuries per 100 million work hours. As far as the total incident trends are concerned for ADNOC, the FPIR is same as 4 in 2020 and 2024.

The lost time injuries rate dropped from 46 to 28 and there is a huge reduction in the recordable injuries rate which dropped from 125 to 39 per 100 million work hours.

When severity is not considered in injury then the values of recordable injuries rates will be subjective and may not be correct in assessing the exact impact on safety as minor injuries of slight, trips, falls, also have same amount of weightage.

The weightage given as per the severity, coupled with frequency gives a better understanding of the matrix and metric for measurement of safety performance. The FPI GATE mechanism takes these things into consideration. In this weightage is given for each occurrence of incident such as first incident, subsequent repetition and so on basing upon the company under observation. This is for both PIs and fatalities separately.

Like the safety performance above, indicators are used to assess the true safety performance using FPIR benchmark. The analysis of each company is tabulated for above. For ADNOC, the FPI performance indicator stood at 0.34 with 2 fatalities and 1 permanent injury case in case of upstream directorate of ADNOC. For other FPI simulations as enumerated above, the FPI score is tabulated.

The FPI simulation data for various ADNOC Group Companies, Business Line Directorates (BLD) were analyzed and the corresponding FPI scores are tabulated for evaluation.

Table 2—ADNOC & IOGP FPIR DATA

ADNOC & IOGP FPIR Data			
	2022	2023	2024
ADNOC Group	1.40	0.90	0.62
BLD US	1.36	0.97	0.89
BLD DS	1.60	0.89	0.35
2023 IOGP - FPIR (>50 MMHRs)	1.40	1.40	1.40
2023 IOGP - 1st Quartile FPIR (>50 MMHRs)	0.60	0.60	0.60
2023 IOGP FPIR	1.67	1.67	1.67

International Benchmarking: IOCS' Approach to FPI and Serious Injury Prevention

Recent sustainability reports of leading international oil companies (IOCs) further validate the emerging industry focus on Fatalities and Permanent Impairments as a more meaningful benchmark compared to traditional frequency-based indicators such as TRIR.

- BP: Focused on Serious Injury and Fatality (SIF) potential methodology, enabling better visibility into serious incident precursors despite a relatively stable TRIR
- Shell: Implemented Targeted Process Safety Fundamentals (PSF) application, explicitly linking critical barrier failures to fatal outcomes
- Chevron: Integrated SIF prevention in Operational Excellence systems, but highlighted divergence between TRIR and FPI. Despite recording continuous improvements in TRIR, Chevron's persistent FPIR underscores the inadequacy of frequency-based metrics to capture and prevent high-severity events.
- TotalEnergies: Shifted to 'Serious Incidents Rate' framework, prioritizing prevention of life-altering events rather than minor recordables.
- ExxonMobil: Deployed Safety Leadership Framework emphasizes serious injury and fatality prevention as core leadership accountable metric.

Across the board, IOCs are converging towards a new paradigm where traditional injury rates (TRIR, LTIFR) are no longer celebrated in isolation but evaluated alongside serious incident prevention success. This reflects the recognition that a declining TRIR does not necessarily correlate with reductions in high-severity events, reinforcing the industry's shift towards FPI/SIF-based performance management.

In this context, ADNOC's pivot toward emphasizing Fatalities and Permanent Impairments aligns with global leaders, reinforcing the urgent need for a standardized international movement towards outcome-based benchmarking that drives prevention of high-consequence events.

ADNOC Performance Highlights (2020-2024)

ADNOC's performance showcases a substantial reduction in TRIR to 0.11 by 2024. Despite this, the number of fatalities and permanent impairments remained volatile. This inconsistency underlines the need to focus on the prevention of high-consequence events rather than only minor injuries.

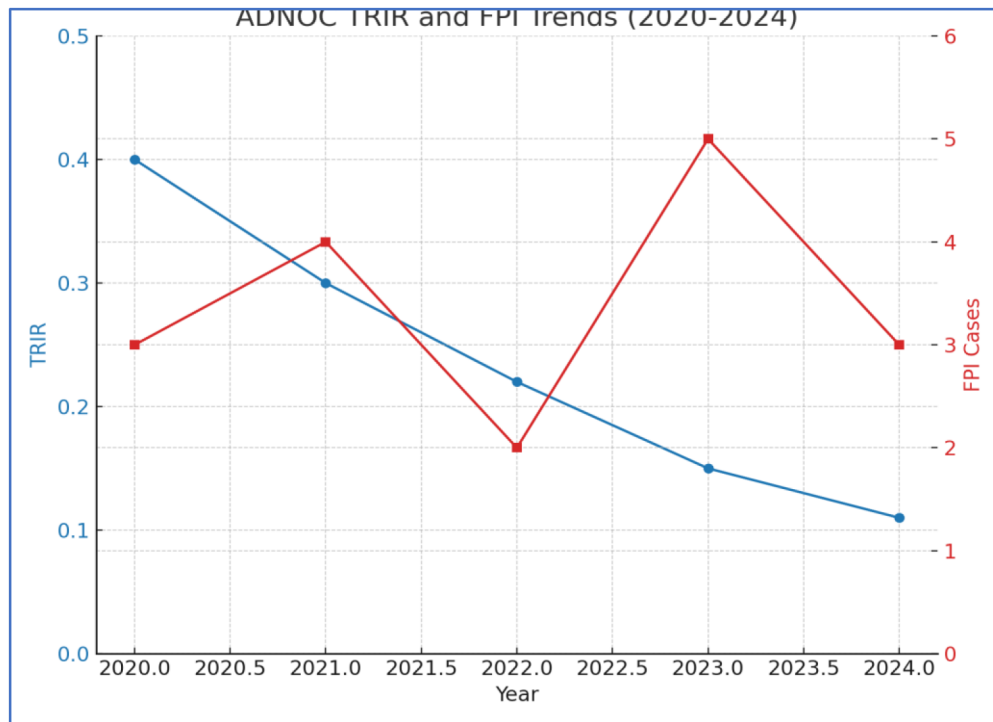


Figure 4—ADNOC TRIR & FPI Trends Yearly

Why TRIR is no Longer Enough

TRIR improvements primarily capture reductions in minor injuries. By contrast, Fatalities and Permanent Impairments directly reflect catastrophic failure of critical barriers. Organizations must therefore prioritize metrics that capture severity, not just frequency. In developing the Risk analysis Matrix (RAM) also it is industry practice to develop RAM based on the product of severity and frequency. The injury occurs based on the risk involved in it. Risk is defined as the product of frequency and severity.

For example, a very high severity incident even if it occurs with less frequency, it carries more risk involved in it than the high frequency low severity incident. Hence ignoring severity of the impact of the incident is not a good industry practice to adopt. The industry safety principles and policies have to be adopted as per the risk matrix rather than considering only frequency. The calculations made in the FPI simulation in ADNOC simulation models do take care of this principle and they are ideal for proper safety assessment.

Hence severity base benchmarks such as FPI benchmark are better for true safety assessment.

The Case for FPI as the Primary Benchmark

Fatalities and Permanent Impairments are unambiguous indicators of serious safety failures. Adopting FPIR as a primary KPI will shift attention towards systemic and critical controls, enhancing true resilience.

Proposed Framework: Integrating FPI Metrics into Safety Management

1. Integrate FPIR alongside TRIR as leading KPIs.

Implementation: Establish FPIR (Fatality Permanent Impairment Rate) as parallel reporting metric to TRIR at both corporate and asset levels. Ensure dashboard, management reviews, and contractor scorecards highlight FPIR trends alongside TRIR, shifting leadership focus from frequency-based reporting to outcome-based performance. This integration should be mandated through corporate KPI cascades to ensure alignment across all business units.

2. Conduct systemic SIF (Serious Injury and Fatality) potential reviews.

Implementation: Institutionalize SIF reviews as part of incident investigations and proactive risk assessments. Use structured tools such as SIF Bowties and barrier analysis to determine whether activities carry FPI potential. Embed this into Management of Change (MoC), risk assessments and audit processes so that SIF potential is routinely considered and tracked, even for events without actual severe outcomes.

3. Enhance barrier and critical control verification linked to FPI risks.

Implementation: Move beyond compliance audits and establish a systematic critical control verification program tied specifically to FPI exposures. Require frontline supervisors and independent assurance teams to verify that life-critical barriers (e.g., process safety valves, confined space permits, fall protection systems) are not only present but effective. Verification outcomes should feed into FPI dashboards for real-time visibility.

4. Embed FPI focuses on contractor and asset lifecycle management.

Implementation: Integrate FPI prevention requirements into contractor prequalification, tendering, and ongoing performance reviews. During asset design and operations, engineers and project managers require to demonstrate how FPI exposures have been identified and mitigated across the lifecycle from FEED (front-end engineering design) through operations and decommissioning. Contractors and project teams should be measured on their ability to reduce high-severity exposures, not just meet frequency-based metrics.

5. Drive leadership safety engagements around life-critical controls.

Implementation: Redefine leadership site visits, safety walks, and forums to focus explicitly on life-critical controls linked to FPI prevention rather than generic checklist observations. Provide leaders with structured engagement guides that prompt meaningful dialogue on critical barriers, SIF learnings, and frontline empowerment. Track the quality and impact of these engagements, ensuring leaders visibly reinforce the importance of preventing severe and irreversible harm

Conclusion

The ADNOC and industry-wide experience demonstrates that reducing injuries does not equate to preventing fatalities. Fatalities and Permanent Impairments must become the centerpiece of safety benchmarking to achieve truly safe, resilient, and sustainable operations.

As explained above the amount of risk involved in any process in an industry depends upon the product of severity and frequency. Hence high severity incidents with low frequency also tend to fatality and high risk requiring more attention towards preventing that incident. Hence switching from TRIR to FPIR based safety evaluation with FPIR as benchmark is a better process and to be adopted by all companies across Globe.

Even though the long-practiced method of adopting TRIR method in figures attractive but in reality, it does not form the actual safety performance indicator as it also contains very low risk incidents with same weightage.

The aspect has been effectively adopted by ADNOC in its regular various operation in downstream and upstream assets. The FPI simulation method practiced by ADNOC is scientifically with high standards and is very effective in analyzing the safety practice on actual ground by adopting to FPIR benchmark.

Acknowledgements

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